



Magnet

Deinstallation Manual

5918186-1EN
Revision 4

Language Policy

DOC0371395 - Global Language Procedure

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Chapter 1 Introduction

1.1 Purpose

This *Magnet Deinstallation Manual* (5918186-1EN) describes the GE HealthCare MR magnet deinstallation and moving process for a GE HealthCare system magnet. This includes the site preparation and steps required to shut down and remove a magnet, in addition to preparing for shipment and eventual magnet transportation. For system deinstallation, refer to the *MR System Deinstallation Manual* (5924907-1EN).

Following these instructions will help GE HealthCare customer service engineers and subcontractors do the disassembly and removal of a GE HealthCare system magnet. This does not include harvested parts such as Magnet Rundown Unit (MRU), MM3 or MM4, compressor, gradients, body coil, or side electronics because any parts directly mounted on the magnet will not be removed with the magnet unless noted otherwise in this manual (for example, [6.2 Removing the other magnet components on page 43](#)).

If preparing to shut down and remove a magnet used in 7T systems, contact the vendor Tesla to remove the magnet from the customer site. Tesla will prepare the magnet for shut down, removal, and transportation.



WARNING

SAFETY RISK

Failure to consult GE HealthCare prior to any system level component deinstallation could result in a safety risk.

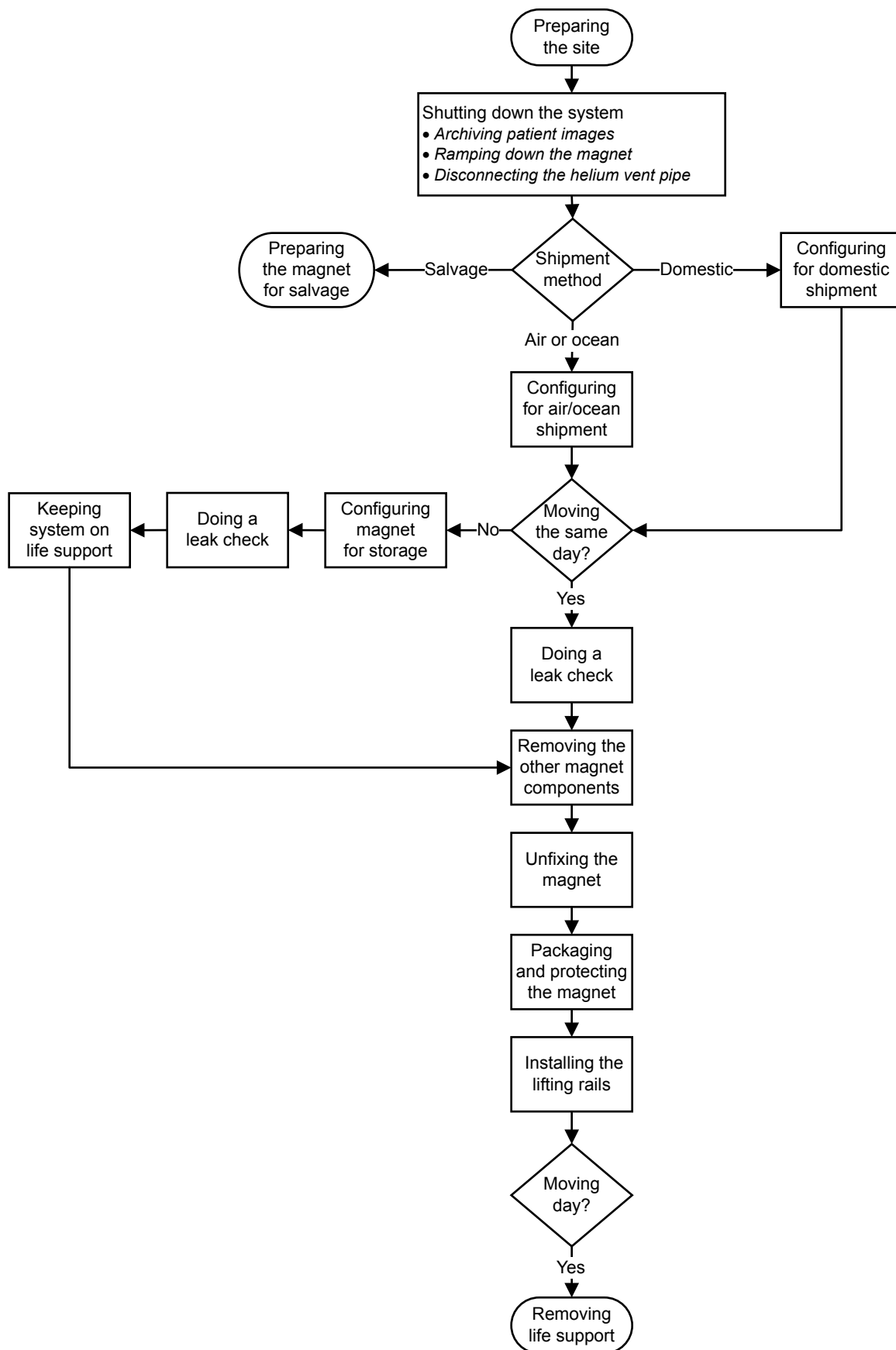
For system-level components (other than the magnet) deinstallation instructions, contact GE HealthCare at 1-800-437-1171 (United States or Canada) or your local service representative.

Reasons for deinstallation

A system must be deinstalled for any of these reasons:

- **Unit sold by customer** - A customer sells or removes a piece of equipment.
- **Unit scrapped or damaged** - A customer or GE HealthCare scraps or destroys a unit.
- **Trade-In** - A unit is traded back in or returned to any GE HealthCare factory, repair, or distribution center.

Magnet deinstallation workflow



1.2 Returning used systems

If you are not returning the system to GE HealthCare for recycling, make sure to comply with local and national regulations for correct disposal and recycling.

1.3 Scope

The information in this manual applies to these magnets:

- 3.0T: AR, UA, and W or WB series
- 1.5T: HM, PM or PC, R or CR, RA, RD, and YR series

Magnets not listed above are **not** included in this manual.

If preparing to shut down and remove a magnet used in 7T systems, contact the vendor Tesla to remove the magnet from the customer site. Tesla will prepare the magnet for shut down, removal, and transportation.

1.4 Definition of terms and acronyms

Term	Definition
GIB	Global Installed Base Database and Application. This is the GE HealthCare database for tracking our products and serial numbers. GIB is the data master for the GE HealthCare service delivery systems.
PMI	GE HealthCare Project Manager of Installation.
Serial Number	A serial number is a unique identifier for traced items. Serial numbers are unique by the model and serial number combination; different models can have the same serial number. A serial number is the only unique identifier that can be traced through the entire manufacturing, shipping, and service systems.
Traced Item	Any product or part that is determined to be traceable is considered a traced item or component. All traced items must have a serial number and labeling corresponding to that serial number. Traced items shall be set up in the respective IB systems as serialized.

1.5 References

For a list of the manuals used to make this *Magnet Deinstallation Manual* (5918186-1EN), see [Appendix A Reference manuals on page 61](#).

1.6 Roles and responsibilities

Table 1-1 Roles and responsibilities

Role	Responsibility
Team roles	Team roles review endorsements and approvals. A team role may be delegated as necessary based on program needs as long as the function originally responsible for the given role delegates authority and the delegate accepts the responsibilities of the given role.
Project manager	A project manager is responsible to make sure all parties are trained and have the required tools, parts, and equipment to complete their tasks safely. Project manager roles at GE HealthCare include Modality Service Leader (MSL) and Project Manager of Installation (PMI). Also can be third party or customer.
Field engineer (FE) and Licensed Channel Partner	FEs and Licensed Channel Partners are responsible for making sure that service records in the Global Installed Base (GIB) are updated any time a traced component is moved, deinstalled, or reinstalled in the field.
Services quality manager	The services quality manager is responsible for making sure that this manual is followed by all Global GE HealthCare employees as well as all other contractors who move or deinstall GE HealthCare magnets and equipment.
Third party vendors	Third party vendors do not have access to all of the GE HealthCare manuals. They are responsible for contacting GE HealthCare for the information to which they do not have access. For example, the information noted in Chapter 4 Shutting down the system on page 21 and Chapter 6 Preparing the magnet for shipment on page 31 .

Table 1-2 Deinstallation functions and responsibilities

Function	Responsibility
Introduction	Project manager, FE, third party vendor
Safety	Project manager, FE, third party vendor
Preparing the site	Project manager, third party vendor
Shutting down the system	FE, third party vendor
Preparing the magnet for salvage	FE, third party vendor
Preparing the magnet for shipment	FE, third party vendor NOTE FEs or Licensed Channel Partners are not responsible for 6.5 Installing the lifting rails on page 48 .
Moving and transporting the magnet	Third party vendor

Chapter 2 Safety

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.



WARNING

SAFETY RISK

Failure to consult GE HealthCare prior to any system level component deinstallation could result in a safety risk.

For system-level components (other than the magnet) deinstallation instructions, contact GE HealthCare at 1-800-437-1171 (United States or Canada) or your local service representative.

2.1 Tools and equipment

Special tools, equipment, and materials may be required on site and must be ordered in advance. See individual procedures for notes and recommendations. To make sure of maximum safety, inspect all tools and equipment before use to be sure they are in good working order.

Always wear any Personal Protective Equipment (PPE) relevant to any procedures.

2.2 Safety training

Do not do any procedures in this manual without correct training.

- Environmental Health and Safety (EHS)
- Hazardous Chemicals in the Workplace
- Hazardous Waste Awareness

Refer to the *MR Service Safety Manual* (5452735). This manual includes the information below:

- MR Service Safety Manual Introduction
- Before Working in the Magnet Room with a Ramped Magnet
- Safety Procedures for Servicing Highly Ferrous Devices
- Magnet Safety
- Superconducting Magnets
- PET/MR Radiation and Safety Guidelines

- Handling of Ferrous Material
- Magnetic Field Considerations
- Emergency Rampdown of Magnetic Field
- Lockout-Tagout (LOTO) Procedures

The excerpts below highlight warnings, safety, and critical tasks from this manual.

Safety Procedures for Servicing Highly Ferrous Devices

This section introduces an explicit service policy when servicing highly ferrous devices in a magnetic field. Devices such as B0 quench boxes, docks, longitudinal drive motors, blowers, passive shims, and coldheads are examples of highly ferrous devices that fall under this policy.

Make sure the *Highly Ferrous Material Pre-Work Checklist* in this section is completed to make sure that the required actions were taken. The second MR trained person also must complete this checklist.

Magnet and Radiation Safety

This chapter includes the topics below:

- Magnet Safety Considerations (for the customer and GE HealthCare)
- Cryogen Safety Requirements
 - Site Precautions
 - Equipment Handling Precautions
 - Cryogen Handling Precautions
 - Magnet Service Precautions
 - First Aid/Emergency Situations
- Magnet Field Safety Requirements

Handling of Ferrous Material

This procedure describes the handling of ferrous objects in the magnet room when various conditions exist.

Emergency Rampdown of Magnetic Field



WARNING

ASPHYXIATION HAZARD!

Emergency magnet ramp down will cause the magnet to quench. If the venting system fails, the magnet room may fill with cryogen gas.

Before doing an emergency magnet ramp down, make sure all room ventilation systems are on and functioning. Make sure the magnet room door is propped open (or the hatch in a mobile van). After initiating emergency magnet ramp down, exit the room immediately and remain near the floor, where oxygen is present. Stay outside of the magnet room until the quench is over and safe room oxygen levels are present.

You can also find this procedure in the appropriate *Magnet and Cryogen Manual* for your magnet series.

Chapter 3 Preparing the site

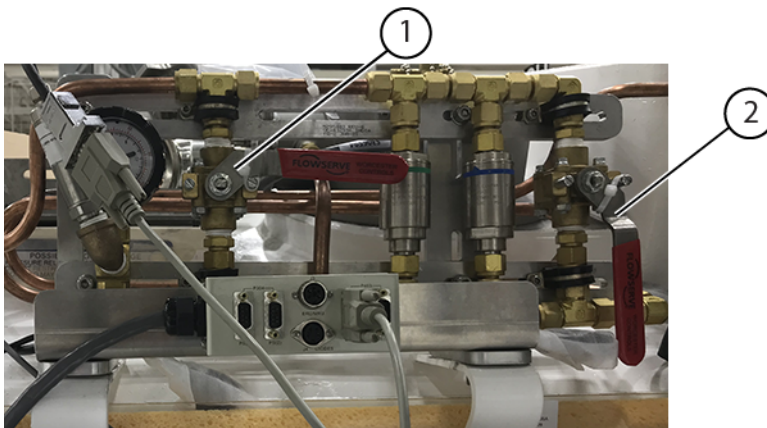
Complete the tasks below before the day you plan to remove the magnet.

1. Identify the route that will be used to remove the magnet, both inside and outside of the facility.
2. Get any permits needed for building construction and road closures.
3. Get the correct magnet handling equipment (for example, crane, lifts, dollies, and so on).
4. Schedule downtime with the facility.
5. Order special tools and equipment as needed.
6. Make sure that helium can flow through both the V2 and V3 (or 0.5 inch NPT plug) valves.

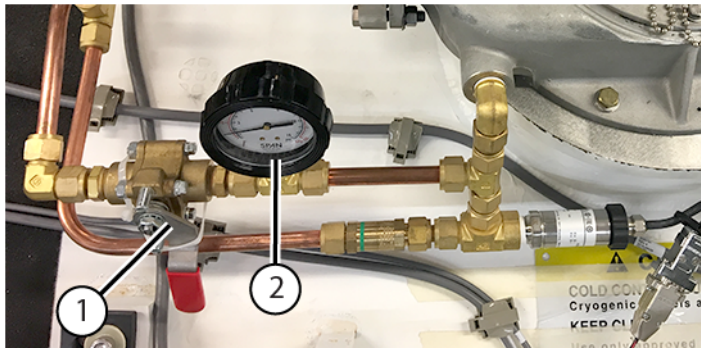
NOTE

If there is no helium flow, contact your local service representative or Florence Engineering.

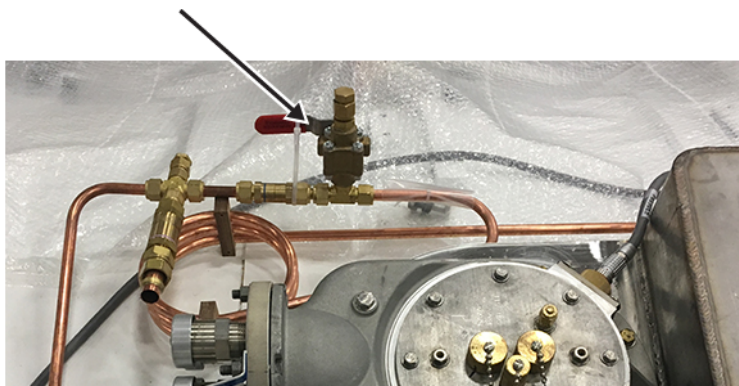
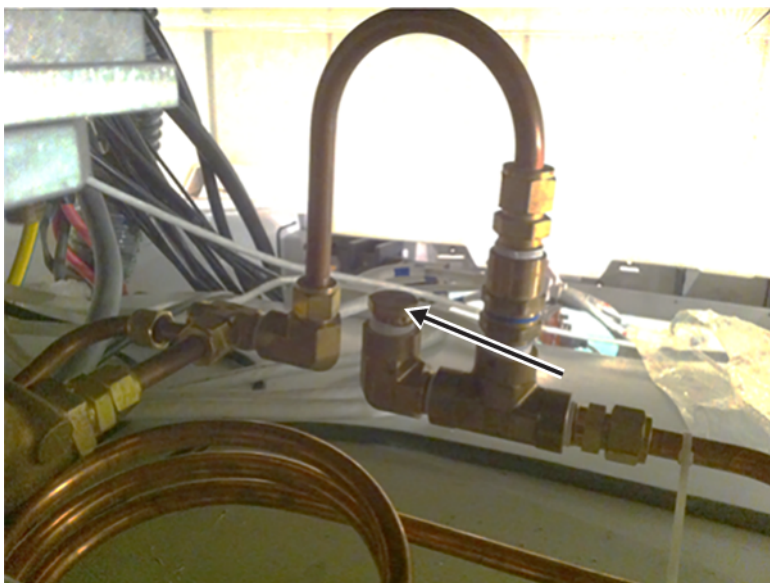
Figure 3-1 AR/PM/PC V2 and V3 vent valves



1	V2 vent valve
2	V3 vent valve

Figure 3-2 Legacy V2 vent valve and pressure gauge

1	Helium vent valve V2
2	Cryostat pressure gauge

Figure 3-3 Legacy V3 vent valve**Figure 3-4 Legacy NPT plug**

7. Visually examine the quench pipe in the scan room and make sure it is in good condition.
8. Look at the quench pipe from outside the building, if possible, and make sure there are no obstructions or objects close to it.

3.1 Equipment inventory

Begin by locating deinstallation equipment and make sure all necessary equipment is on-site before beginning deinstallation. If equipment is missing, take appropriate steps to get these items.

3.2 Floor protection

The deinstallation team is required to provide floor protection in the room where the equipment is being deinstalled. The movers are responsible for floor protection from the scan room to the loading area of their truck.

3.3 Equipment handling

NOTICE



POSSIBLE COMPONENT DAMAGE

A drop from a height greater than 12 mm (0.50 inches) induces structural damage to the frame or other major components. This damage is not always obvious or clearly visible.

Never drop subsystem components (system cabinets, chillers, patient table, rear pedestal, GOC). All system components must remain upright at all times, and must not be tipped. The project manager should contact engineering for all special lifting requirements.

For door opening and elevator requirements, refer to the appropriate *Pre-Installation Manual* to identify equipment specifications, dimensions, and weight.

3.4 Health Insurance Portability and Accountability Act (HIPAA)

In the USA, the Health and Human Services (HHS) department has established a set of standards entitled Health Insurance Portability and Accountability Act (HIPAA). A key component of these standards are the laws and ensuing regulations, which established guidelines for individual patient management and the access to information. Included is the provision of a legal firewall to prevent inappropriate, unauthorized disclosure of patient information. This applies to access of confidential electronic health information found on the console computer and hard drives, PACC systems, and Advantage Workstations.

3.4.1 Patient privacy notice

Laws and regulations are established and globally enforced in all countries for patient privacy management and unauthorized access to patient information. In an effort to prevent inadvertent disclosures of patient information during a system deinstallation, GE HealthCare and/or their agents shall remove all patient data, patient images, and image raw data prior to system final power down at deinstallation. Procedures in this manual represent the minimum tasks required to meet global patient privacy standards, acts, and regulations. If your country, state, region, or local laws regarding patient and privacy standards are more stringent than stated in this manual, always follow the requirements for your area:

- Image compute nodes
- Console computers and hard drives
- PACC systems and storage systems
- Advantage Workstations and storage systems

3.5 Shipping and removal

Important

All shipping dimensions and weights are approximate and may vary based on ship-to location, required rigging, or other requirements. Some shipping or access routes may have requirements in addition to those listed in this section. Contact the GE HealthCare Project Manager of Installation (PMI) to make sure of magnet shipping, rigging, and access.

3.5.1 Cable removal requirements

Some cables, water hoses, and helium pressure lines are intended to be reused. Do not use excessive force when pulling cables to avoid damage to connectors. Take special care when handling fiber optic and RF cables. Avoid twisting, squeezing, pinching, or bending the fiber optic cable beyond the specification for fiber optic cable.

Before disconnecting cables, examine the labels. Reattach any loose labels (for example, with clear Scotch tape); replace missing labels (handwritten using permanent marker on adhesive tape).

All items must be packed properly to avoid damage during transportation.

NOTE

Do not put heavy gradient cables on top of fragile signal cables or fiber optics.

When wrapping cables, avoid any tangles and knots. Use cable ties to secure the cables together.

Make sure that cables, water hoses, and helium pressure lines are protected and secure for transportation.

3.5.2 Facility removal route requirements

Refer to the *MR System Deinstallation Manual* (5924907-1EN) appendix for the appropriate service documentation listing the removal dimensions of system components. Upon removal, confirm the component dimensions and weight. The removal route must be planned to accommodate the dimensions listed.

NOTE

Avoid blocking emergency exits or other vital access routes in the hospital. Do not damage walls, doors, or other parts of the building during the deinstallation process.

NOTE

The customer must provide an area for removing system components from the MR Suite and loading them into the truck. For appropriate magnet handling and removal documentation, see [Chapter 7 Moving and transporting the magnet on page 59](#).

3.5.3 Shipping and removal safety



SAFETY HAZARD

All safety precautions **MUST BE FOLLOWED!** Failure to do so may result in equipment damage, death, or serious physical injury! Pay attention to all safety notes.

3.5.3.1 Equipment damage risk

Improper transportation can result in damage to the equipment. Review guidelines with the carrier prior to transporting the equipment. Any other shipping configuration must be demonstrated through testing and be approved by GE HealthCare.

Impacts/jolts/drops to the equipment while lifting/moving/lowering the equipment can cause expensive damage. The rigger is responsible for actual equipment/procedures used to lift and move equipment out of the customer facility, including through a raised opening in an exterior wall. Lift/move/lower the equipment smoothly. Do not let the equipment bump or hit anything forcefully. Avoid tilting the equipment; all equipment should remain in an upright position. Do not apply any force to the equipment enclosures. Protect all customer surfaces during any and all move operations.

All subsystems must be protected from the weather. Make sure appropriate packing material is placed so they will not be exposed to water damage or dust contamination. For temperature and humidity storage requirements, refer to the appropriate *Pre-installation Manual*.

3.5.3.2 Potential injury risk

Magnet and subsystems are unbalanced loads. Tilting can result in a hazardous condition that can cause serious personal injury.

To avoid personal injury **NEVER** do the following:

- Stand under heavy loads
- Use defective or damaged tools or other equipment
- Reuse mounting hardware when lifting the magnet
- Overtighten nuts or bolts above the recommended torque specification
- Let unauthorized or authorized personnel stay close to a moving load
- Exceed the load capacity of devices
- Route or stretch ropes, chain, and straps over sharp edges
- Knot rope or chains
- Twist ropes
- Use compressible materials for floor protection (plastic mats or similar), as this creates instability when moving heavy parts
- Stand between a moving part and a stationary object

ALWAYS do the following:

- Wear personal protective equipment

- Make sure no screws protrude below the pallet (not possible to put the forks of the forklift below the pallet)
- Be aware of pinch and crush points, especially for hands and feet
- Wedge or apply brakes to equipment with wheels, to prevent unintended motion
- Clear route before moving equipment to avoid tripping and beware of surroundings
- Watch for possible risks of lacerations (such as from cable tie wraps, sharp metal corners, and so on)
- Clean up spills to avoid slipping
- If possible, avoid lifting heavy equipment more than 45 mm above the floor
- Avoid awkward lifting positions
- Avoid biohazards and make sure they are properly cleaned up before packaging equipment
- Make sure there are qualified spotters present when moving equipment
- Make sure each subcomponent is securely fixed to pallets and crates
- Make sure each subcomponent is securely fixed inside of the truck or other form of transportation (use straps if necessary)
- Make sure the magnetic field is zero prior to shipping

NOTICE

Before beginning the deinstallation process, make sure the magnet has been properly ramped down. If not, refer to *Shutting down the system* in the *Magnet Deinstallation Manual* (5918186-1EN).

Chapter 4 Shutting down the system

Important

The procedures in this chapter must be done by a trained GE HealthCare service engineer or a trained Licensed Channel Partner.

4.1 Archiving patient images

1. Follow the processes outlined in the *MR System Deinstallation Manual* (5924907-1EN) for archiving patient images and shutting down the rest of the system.

4.2 Ramping down a magnet to zero field

Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Ramping down a magnet to zero field
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Replacement and Maintenance > Ramping down a magnet to zero field
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Magnet Rampdown to Zero Field
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Replacement and Maintenance > Ramping down a magnet to zero field
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Replacement and Maintenance > Ramping down a magnet to zero field
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Replacement and Maintenance > Ramping down a magnet to zero field
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Replacement and Maintenance > Ramping down a magnet to zero field

4.3 Removing the field from the magnet

4.3.1 Normal ramp down

The manuals listed in [4.2 Ramping down a magnet to zero field on page 21](#) need to be followed when ramping down a magnet. The procedures for ramping down listed in these manuals state that the current in the magnet must reach 0A and the main coil voltage should reach 0V on the main power supply.

NOTE

Here and below 0V is considered as either equal to 0V on the power supply main coil voltage meter, or the voltage did not decrease any lower for at least 15 minutes since the main coil current reached 0A.

1. Wait for 10 more minutes and then turn the main and axial heaters off.
2. **(For R, CR, RA, YR, W, and WB series magnets)** Connect the heater cable to the shim power supply, turn the Axial, Transverse 1 (T1), and Transverse 2 (T2) heaters on, and wait a minimum of 5 minutes. Then turn the Axial, T1, and T2 heaters off.
3. Follow all air instructions for air shipment as required. See [4.3.4 Air shipment on page 23](#).

4.3.2 Ramping down with the auxiliary cable due to significant icing in the vertical penetration (only R, CR, RA, YR, W, and WB series)

If the shim lead is unable to make a connection due to significant ice in the vertical penetration, and the auxiliary cable is used to ramp the magnet down, after the ramp down is completed, follow the steps below.

1. Inject 99.999% helium gas into the shim lead and seat the shim lead correctly so continuity is established. Refer to the **Replacement and Maintenance > Deicing the magnet > Deicing the shim lead and SavCon connector > Deicing the shim lead and SavCon connector level 1** procedure in the appropriate *Magnet and Cryogen Manual*:
 - **(For R, CR, RA, or YR series magnets)** *Magnet and Cryogen Manual for 1.5T R Series Magnets* (5928384-8EN).
 - **(For W or WB series magnets)** *Magnet and Cryogen Manual for 3.0T W or WB Series Magnets* (5928479-8EN).
2. When the shim lead has continuity, then use the normal main power supply and open the axial and main heater switches.
3. Make sure main coil voltage and current on the main power supply are zero.
4. Wait for 10 more minutes and then turn the main and axial heaters off.
5. Connect the heater cable to the shim power supply, turn the Axial, Transverse 1 (T1), and Transverse 2 (T2) heaters on, and wait a minimum of 5 minutes. Then turn the Axial, T1, and T2 heaters off.
6. If air shipment is needed, follow all air instructions for air shipment as required. See [4.3.4 Air shipment on page 23](#).

4.3.3 Ramping down with the auxiliary cable due to SavCon or shim lead damage (only R, CR, RA, YR, W, and WB series)

If the shim lead is unable to make a connection due to faulty hardware, and the auxiliary cable is used, follow the steps below.

1. Make sure main coil current on the main power supply is zero. There are no voltage tabs to measure main coil voltage.
2. After the magnet is ramped down using the auxiliary cable, fix the hardware problem and reestablish the shim lead connectivity.
 - If shim lead connectivity cannot be reestablished, turn the compressor off and let the magnet warm up above 10 K (per cryostat diodes) to let all superconductive coils quench.
 - If shim lead connectivity is reestablished, continue to [Step 3](#).
3. When the shim lead has continuity, then use the normal main power supply and open the axial and main heater switches.
4. Make sure main coil voltage and current on the main power supply are zero.
5. Wait for 10 more minutes and then turn the main and axial heaters off.
6. Connect the heater cable to the shim power supply, turn the Axial, Transverse 1 (T1), and Transverse 2 (T2) heaters on, and wait a minimum of 5 minutes. Then turn the Axial, T1, and T2 heaters off.
7. If air shipment is needed, follow all air instructions for air shipment as required. See [4.3.4 Air shipment on page 23](#).

4.3.4 Air shipment

Follow International Air Transport Association (IATA) specifications for air shipment. If you cannot confirm that you meet the requirements, the magnet must be warmed up to 10 K.

4.4 Disconnecting the helium vent pipe

Required conditions

For Ventglas removal, the magnet must be ramped down.

1. Set up the service platform. Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Installation and Adjustment > Setting up the service platform
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Installation and Adjustment > Setting up the service platform
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Service Platform Set-Up

Magnet series	Manual title	Part number	Procedure
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Installation and Adjustment > Setting up the service platform
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Installation and Adjustment > Setting up the service platform
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Installation and Adjustment > Setting up the service platform
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Installation and Adjustment > Setting up the service platform

2. Make sure that the magnet is ramped down.
3. Remove the insulation from around the exhaust gas vent pipe from the base of the vent (at the magnet) to above the Ventglas fiberglass wrap.
4. Remove tape from both ends of the Ventglas, and any tape over the gap between the vent pipe and the ceiling vent pipe.

NOTE

The Ventglas can be either vertical and directly above the magnet vent pipe or horizontal and just after the 90-degree elbow that is directly connected to the magnet flange.

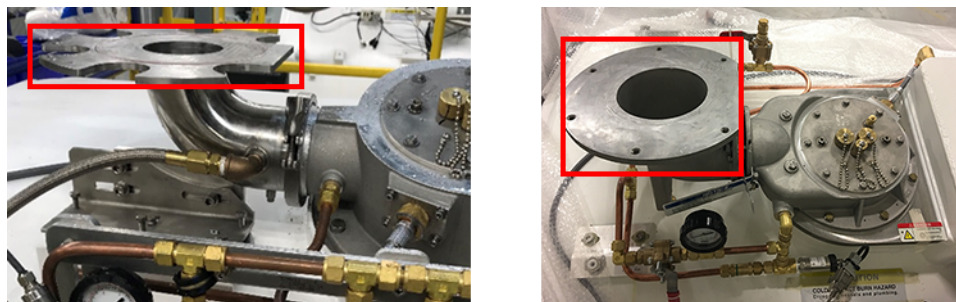
5. Remove the Ventglas from the vent pipe. Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Cryogen Exhaust Gas Vent Connection
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas

Magnet series	Manual title	Part number	Procedure
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Replacement and Maintenance > Replacing the Ventglas on the exhaust gas vent pipe > Removing the Ventglas

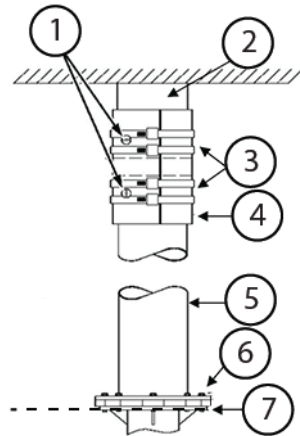
6. Disconnect and remove the exhaust gas vent pipe.
 - a. Loosen the six mounting cap screws and washers that connect the vent pipe to the vent adapter flange or 90-degree elbow. Do **not** remove the vent adapter flange, which is connected to the magnet.

Figure 4-1 Vent adapter flange with vent pipe removed - AR/PM/PC magnet (left) and legacy magnet (right)



- b. Remove all hardware, including the clamp ring, O-ring, vent pipe, and any vent pipe 90-degree elbows from the top of the magnet flange (in other words, everything above the dotted line in [Figure 4-2 Helium vent kit parts and assembly](#)).

Figure 4-2 Helium vent kit parts and assembly



1	No. 10 - 16 x 1.0 inch screws (2)
2	Ceiling vent pipe and RF penetration waveguide
3	Hose clamps (4)
4	Ventglas fiberglass connector sleeve
5	Vent pipe
6	Bolts or cap screws (6), depending on the magnet series, and washers (6)
7	Vent adapter flange

- c. Put the removed hardware in the box for shipment with the magnet.

NOTE

If the magnet will be left in this state for more than 24 hours before removal, the emergency exhaust vent must be turned on, or it must be wired to an operational oxygen monitor so that it starts automatically if oxygen levels in the magnet room are lower than normal.

4.5 Cryogen and magnet life support

Leave the magnet connected to the cryocooler and monitoring system (MM3 or MM4) to keep correct temperature and pressure, and to keep helium loss and internal icing issues at a minimum. Refer to the *Magmon3 Installation and Service Manual* (5124576) or *Magnet Monitor 4 Operating and Service Manual* (5804162) and the appropriate *Magnet Handling Manual* for your magnet series (see [Appendix A Reference manuals on page 61](#)).

These manuals include the information below:

- Connecting the magnet to the compressor
- Connecting the magnet monitoring connections
- Connecting the warehouse magnet monitoring connections
- Monitoring the magnet
- Monitoring the magnet at the warehouse

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Chapter 5 Preparing the magnet for salvage

Superconducting magnets that are serviced by GE HealthCare, being removed from customer sites, and sent to salvage require special consideration when planning their decommissioning, especially for removal of the cryogenics.

Use the steps below, along with the appropriate *Magnet Handling Manual* for your magnet series, when decommissioning before shipping to salvage.

1. Review the *MR Service Safety Manual* (5452735) and follow established safety procedures (for example, the buddy system).
2. Before taking the magnetic field down, make sure the helium exhaust vent is intact and contiguous to the building exit, and the Emergency Rundown Unit (ERU) / Magnet Rundown Unit (MRU) is fully functional and connected to the magnet.
3. The magnet must be ramped down to zero field. Follow the procedure documented in the appropriate *Magnet and Cryogen Manual* for your magnet series.
4. Before shipping to salvage, the magnet must be warmed up, pressure at 0.0 psi, and the main cryostat vacuum brought up to atmospheric pressure in a controlled manner. Contact Florence Engineering for the approved process used for the magnet.

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Chapter 6 Preparing the magnet for shipment

Important

The procedures in this chapter must be done by a trained GE HealthCare service engineer or a trained Licensed Channel Partner.

Required conditions

The magnet must be at zero field before starting this chapter.

Safety

NOTICE

POSSIBLE COMPONENT DAMAGE

Personnel must take care when deinstalling a magnet to prevent damage or loss of components. Failure to do so may result in safety concerns as well as unnecessary delays and expense.

6.1 Converting a magnet to shipment configuration

Safety

Before working in any GE HealthCare MR suite or doing any GE HealthCare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE HealthCare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE HealthCare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

DANGER



POTENTIAL ASPHYXIATION

Large quantities of gaseous helium are discharged from the magnet during installation and commissioning activities. Gaseous helium is an invisible, odorless gas, which will displace oxygen in the air and potentially cause asphyxiation.

Before bringing the magnet into the magnet room, make sure that both the magnet room and magnet venting systems are installed in conformance with the applicable preinstallation manual's cryogenic venting and room venting sections. Proper exhausting will prevent oxygen displacement in the magnet room.

Safety	
	⚠ WARNING POTENTIAL INJURY If you stand directly on top of the magnet, you risk falling and possible injury. Do not stand on top of the magnet! Do all service actions that require access to the top of the magnet from the service platform supplied with the magnet.
	⚠ CAUTION POTENTIAL COLD BURN Exposure to cryogenics and contact with cold objects is possible during this procedure. Wear protective clothing, nonabsorbent gloves, and goggles while doing this procedure.
	⚠ CAUTION POTENTIAL PINCH HAZARD Fingers placed in the path of moving components while the service platform is being assembled or adjusted can be pinched. Do not put your fingers in the path of any service platform components that are being assembled or adjusted.
	⚠ CAUTION POTENTIAL CUT HAZARD The service platform may have sharp edges that can result in injury. Wear leather gloves while assembling, adjusting, or disassembling the service platform to avoid getting any cuts or scratches and to keep your hands clean.
	⚠ CAUTION POTENTIAL BACK INJURY Poor posture when lifting can cause back strain or damage. Keep the upper body directly over your legs and use the service platform's lifting handles when lifting an end of the platform.

When on-site, the magnet is in an operating and site configuration. To prepare for a domestic (ground), air, or ocean shipment, you must first convert the magnet to a shipping configuration.

NOTE

This procedure qualifies the magnet for ground transport. For air shipment of a magnet, additional demagnetization checks must occur. Contact your GE HealthCare representative for instructions.

6.1.1 Configuring a legacy magnet for shipment

The procedures in this section outline how to convert legacy magnets (HM, R, CR, RA, YR, RD, UA, W, and WB) to a shipping configuration.

6.1.1.1 Converting an air or ocean shipment

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Nonmagnetic Titanium Service Tool Kit, Large Set • 9/16 Inch Wrench • Ratchet • Extension • 7/16 Inch Socket	1	5112581	-

Consumables			
Item	Quantity	Part number	Manufacturer
Cable Tie Wraps	1	46-208758P3	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
Blanking Plate Assembly, Four Valve, DVw_3TLC	1	5418413	-

1. Get the four-valve blanking plate assembly (5418413). You may need to contact Florence or Tianjin.
2. Open the V2 and V3/NPT vent valves to make sure there is gas flow through the thermal shield line.

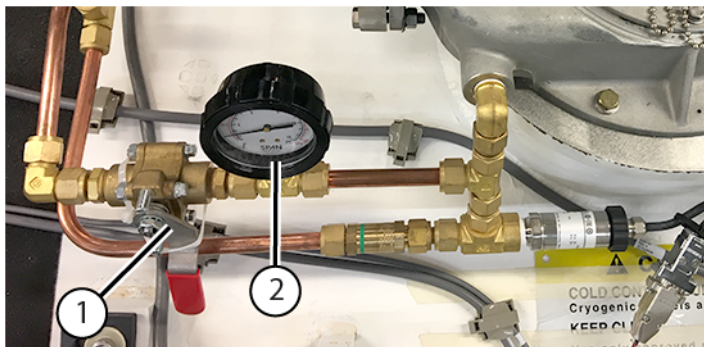
NOTE

If there is no helium flow, contact your local service representative or Florence Engineering.

3. Look at the pressure gauge and make sure that the magnet has positive pressure, between 0.2 to 0.5 psig.

NOTE

Minimum positive magnet pressure is required to keep air from entering the magnet and creating ice.

Figure 6-1 Legacy V2 vent valve and pressure gauge

1	Helium vent valve V2
2	Cryostat pressure gauge

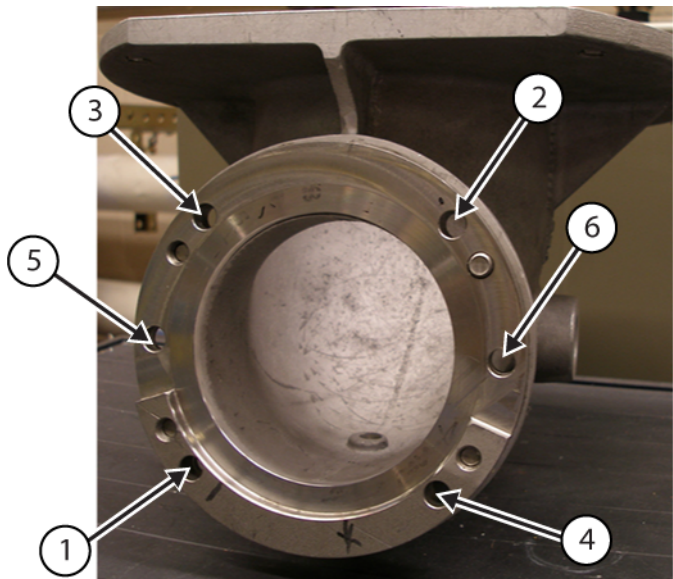
4. If you have plumbing assembly 5555227, you must use vent valve V3 to vent down to < 0.2 psig. Vent valve V2 stops venting at approximately 0.5 psig.
5. Disconnect the plumbing from the 0.5 psi check valve, which is on the rear side of the vent adapter.

Figure 6-2 0.5 psi check valve (disconnect point)

6. Turn off the compressor when doing the actual task.

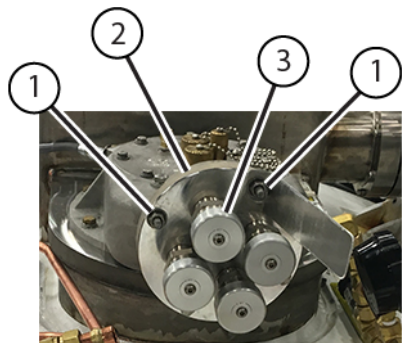
- 7. Loosen and remove all six retaining bolts and corresponding hardware from the plenum assembly. This includes removal of the vent adapter, burst disc, and caution label.

Figure 6-3 Retaining bolts



- 8. Reuse four of the removed retaining bolts to install the blanking plate and caution label.

Figure 6-4 Installing the blanking plate



1	Retaining bolt (4)
2	Blanking plate
3	15 psig relief valve (4)

9. Put all remaining hardware in a plastic bag and secure it to the plumbing assembly with a tie wrap. Make sure the contents of the bag includes two hex bolts, two nuts, two flat washers, and two Belleville washers.

Figure 6-5 Hardware bag secured to plumbing assembly



10. Examine all plumbing for a correct fit and tight connectors.
11. Turn the compressor back on.
12. Put the burst disc, gasket, and vent adapter in a box and write the magnet serial number on the outside of the box.

Figure 6-6 Correct packaging of burst disc



6.1.1.2 Converting a domestic shipment

1. Make sure a good, undamaged burst disc is installed. Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Replacement and Maintenance > Replacing a burst disc
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Replacement and Maintenance > Replacing a burst disc

Magnet series	Manual title	Part number	Procedure
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Replacement and Maintenance > Replacing a burst disc
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Replacement and Maintenance > Replacing a burst disc
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Replacement and Maintenance > Replacing a burst disc

- Open the V2 and V3/NPT vent valves to make sure there is gas flow through the thermal shield line.

NOTE

If there is no helium flow, contact your local service representative or Florence Engineering.

6.1.2 Configuring an AR or PM/PC magnet for shipment

The procedures in this section outline how to convert AR or PM/PC series magnets to a shipping configuration.

6.1.2.1 Converting an air or ocean shipment

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Nonmagnetic Titanium Service Tool Kit, Large Set	1	5112581	-

Consumables			
Item	Quantity	Part number	Manufacturer
Cable Tie Wraps	1	46-208758P3	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
4.237 ID x 0.103 THK, O-Ring 2-156 (included in Burst Disc FRU Kit, Platform, 5863474)	1	46-281101P12	-
Gasket, PTFE, Platform (included in Burst Disc FRU Kit, Platform, 5863474)	1	5795501	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
Burst Disc Clamp, 4-Inch	1	5779337 or 5885378 NOTE If you have the older clamp 5779337, reuse it; if a new clamp is needed, order 5885378.	-
International Shipping Plate FRU Kit, Platform	1	5807233	-

1. Look at the pressure gauge and make sure that the magnet has positive pressure, between 0.2 to 0.5 psig. For more information on how AR and PM/PC series absolute magnet pressure is regulated and reported to the Magnet Monitor (MM4), see [Appendix B Magnet pressures – P_{MM4} versus psig on page 63](#).

NOTE

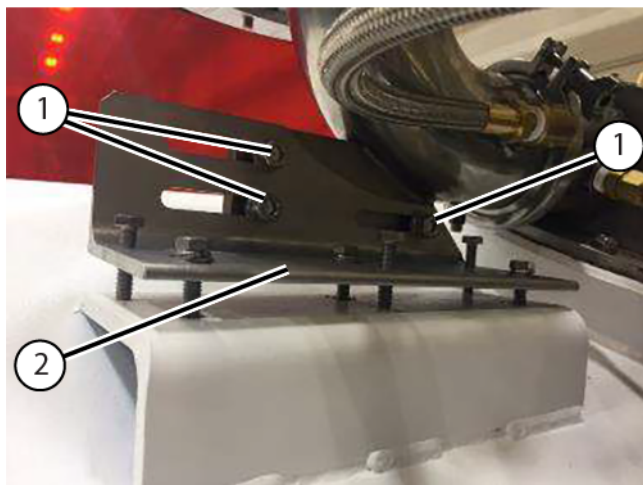
Minimum positive magnet pressure is required to keep air from entering the magnet and creating ice.

2. Turn off the compressor when doing the actual task.
3. Make sure the burst disc is removed. Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Replacing a burst disc > Removing the burst disc
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Burst Disk Replacement

4. Loosen and remove the three M10 bolts and the vent adapter from the vent adapter support bracket.

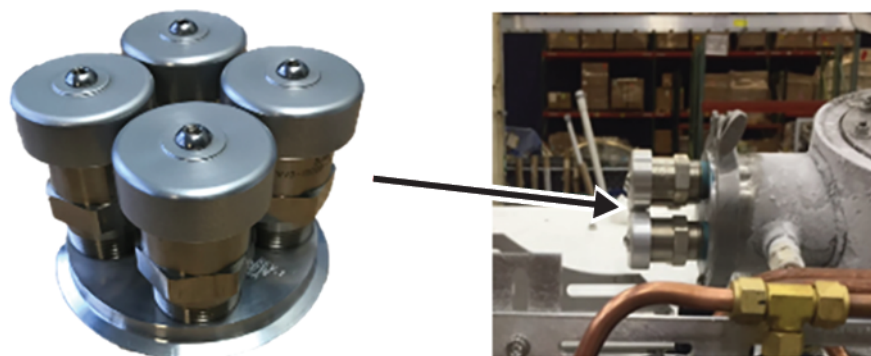
Figure 6-7 Vent adapter support bracket



1	M10 bolts
2	Vent adapter support bracket

5. Remove the M10 bolts and associated M10 hardware below the vent adapter.
6. Get the international shipping plate assembly (FRU 5807233) and install it with a PTFE-encapsulated O-ring (46-281101P12) and 4 inch tri-clamp (5779337 or 5885378).

Figure 6-8 International shipping plate

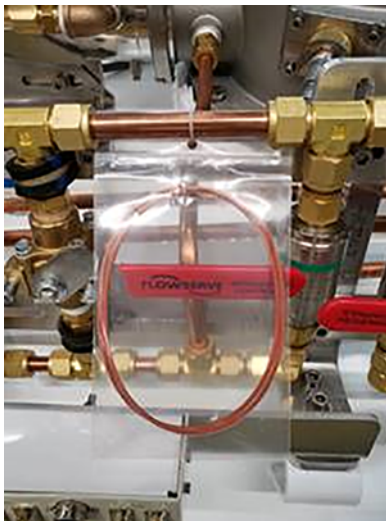


7. Remove the O-ring from the top of the vent adapter, twist the O-ring into an infinity loop, fold it in half at the twist, and put it in a plastic bag. Use a zip tie to secure the bag to the copper tubing of the plumbing assembly.

Figure 6-9 Vent adapter O-ring - installed (left) and removed (right)



Figure 6-10 Vent adapter O-ring ready for shipment



8. Store the vent adapter with the magnet and reinstall the M10 hardware on the vent adapter.
9. Examine all plumbing for correct fit and make sure all connections are tight.
10. Turn the compressor back on.

11. Put the removed burst disc, PTFE sanitary gasket (5795501), and PTFE-encapsulated O-ring (46-281101P12) in a box and write the magnet serial number on the outside of the box.

Figure 6-11 Correct packaging of burst disc



Important

The rating plate must always stay with its respective burst disc. If the metal tab on the burst disc breaks during removal, keep the rating plate with the burst disc so the tab can be attached in the future to the area where the burst disc is installed.

12. Open the V2 and V3 vent valves to make sure there is gas flow through the thermal shield line.

NOTE

If there is no helium flow, contact your local service representative or Florence Engineering.

6.1.2.2 Converting a domestic shipment

1. Make sure a good, undamaged burst disc is installed. Refer to the appropriate manual below for your magnet series.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Replacing a burst disc > Installing the burst disc
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Burst Disk Replacement

2. Open the V2 and V3 vent valves to make sure there is gas flow through the thermal shield line.

NOTE

If there is no helium flow, contact your local service representative or Florence Engineering.

6.1.3 Doing a leak check

Refer to the appropriate manual below for your magnet series to do helium leak troubleshooting.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks
PM or PC	Magnet and Cryogen Manual for Platform Magnets	5796143-4EN	Magnet Functional Checks
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Installation and Adjustment > Doing a system leak test
			Magnet Troubleshooting and Checks > Troubleshooting Leaks

6.2 Removing the other magnet components

Refer to the following procedures in the appropriate system service manual for your product system.

NOTE

The procedures listed below are general. The actual procedure titles and navigation paths to access these procedures within the service manual may differ between systems.

1. Removing the dock and patient table
2. Removing the coolant from magnet components

6.3 Unfixing the magnet from the floor

1. If applicable, remove all magnet anchoring bolts and hardware.
2. If necessary, tape shim edges to avoid injury.



CAUTION

POTENTIAL PERSONAL INJURY

Shims have sharp edges that can cause personal injury.

Always tape shim edges to the floor surface.

3. Except for the compressor and Magnet Monitor cables, disconnect all other power, control, and system level cables attached at the magnet. Refer to the connection block diagrams in the *Installation Manual* for your system, and follow reverse order to disconnect and remove cables that were installed during system installation.

NOTE

Do this step on or the day before the magnet is removed.

4. Remove magnet covers as needed to prepare for lifting rail installation, rigging, and moving. Protect removed covers with bubble wrap or equivalent material for transportation.

6.4 Packaging and protecting the magnet

NOTE

Packaging consumables can be obtained from your local hardware supply store as needed. This includes, but is not limited to: bubble wrap, packaging foam, packing tape, and cable tie wraps.

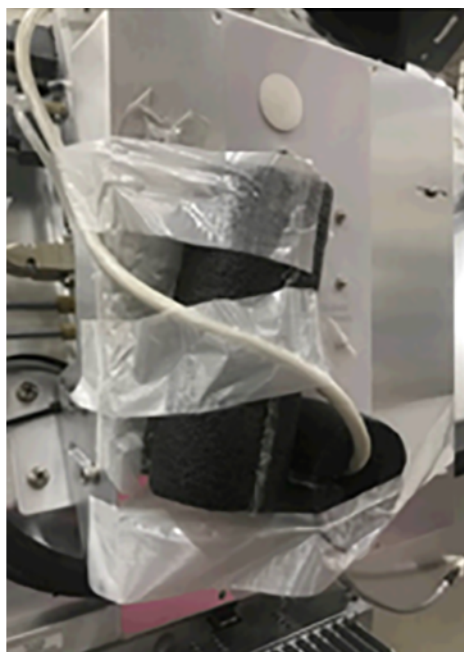
1. Do not cut cables or hoses. The exception is if the hose cannot be removed from the fitting, then cut the hose as close to the fitting as possible.
2. Prevent damage to special cables, if present [for example, In-Room Display (IRD) and fiber optic cables].

Package the IRD cables and bracket as shown in the figure below using bubble wrap or packaging foam and tape. Secure the wrapped cables to the IRD bracket assembly with a tie wrap to limit movement during shipment.

Figure 6-12 Packaging the IRD cables and bracket with bubble wrap**NOTE**

Do not use tie wraps to secure the cables unless they are wrapped in bubble wrap or packaging foam to prevent damage to the fiber optic cable. Also use caution when rolling up any fiber optic cable and do not bend the cable too tightly; use loose turns to prevent damage to the fiber optic cable.

Package the Peripheral Gating (PG) cable where it attaches to the Physiological Acquisition Controller (PAC) as shown below.

Figure 6-13 Packaging the PG cable

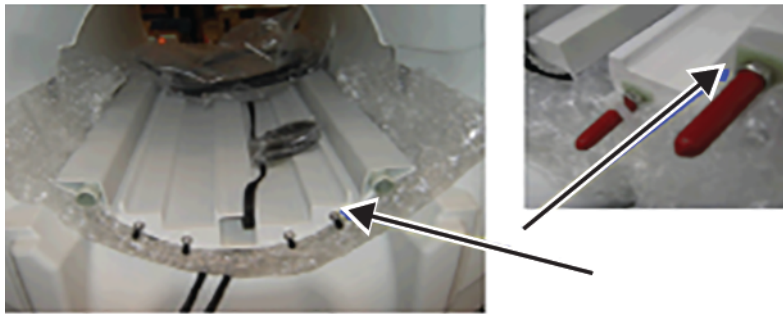
3. Prepare the bridge. Reference the examples below, and refer to the *MR System Deinstallation Manual* (5924907-1EN).

NOTE

Decontamination may be needed before continuing. If you are unsure, contact the customer or local service representative for guidance.

- a. Secure the rear bridge nuts on the bridge bed using fiber reinforced tape (or an equivalent).

Figure 6-14 Rear bridge nuts

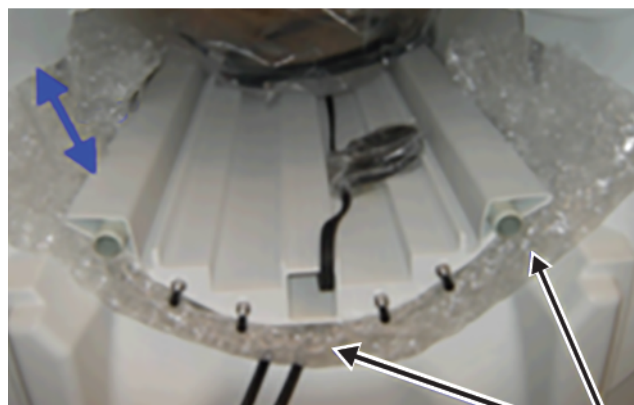


- b. Cut bubble wrap pieces approximately 36 inches x 45 inches long.
 - i. Fold the bubble wrap pieces lengthwise and put a thickness of approximately 3 to 4 inches between the end of the bridge and the end bell.
 - ii. Make sure that the bubble wrap does **not** extend beyond the seam between the end bell and RF coil.
 - iii. Make sure that the bubble wrap **does** extend 2 to 6 inches beyond both bridge sides and the bridge end.

Figure 6-15 Bubble wrap extending 2 to 6 inches beyond bridge side (left arrow span) and not beyond the RF coil seam



Figure 6-16 Bubble wrap extending 2 to 6 inches beyond the bridge side and end



- c. Using a second piece of bubble wrap, make a 12 inch x 12 inch pad that is at least 2 inches thick (approximately). Tape the pad using packaging tape.

NOTE

You can tape multiple pieces of bubble wrap together to create the pad, if needed.

Figure 6-17 12 x 12 bubble wrap pad



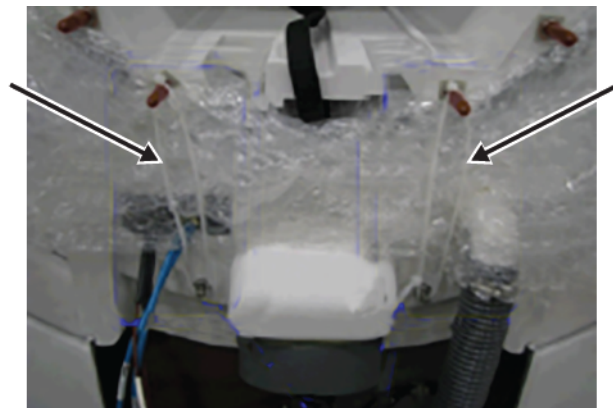
- d. Put the pad at the midline of the bridge at the rear end of the magnet.

Figure 6-18 12 x 12 bubble wrap pad position



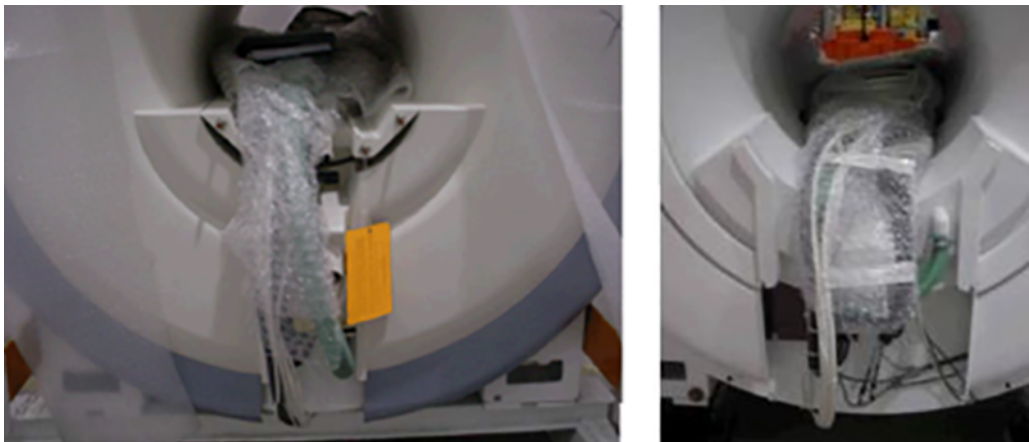
- i. Lift the bridge as needed to insert the bubble wrap pad. Use caution during any bridge movement to make sure cables or adjacent components are not damaged.
 - ii. Make sure that the pad is not touching the RF coil airflow adapters.
- e. Secure the bridge for shipment using a series of cable ties daisy-chained together between the bridge and the vessel or the bridge and adjacent enclosure component screw. The number of locations to secure is different for each system.

Figure 6-19 Cable ties securing the bridge to the cover/end bell



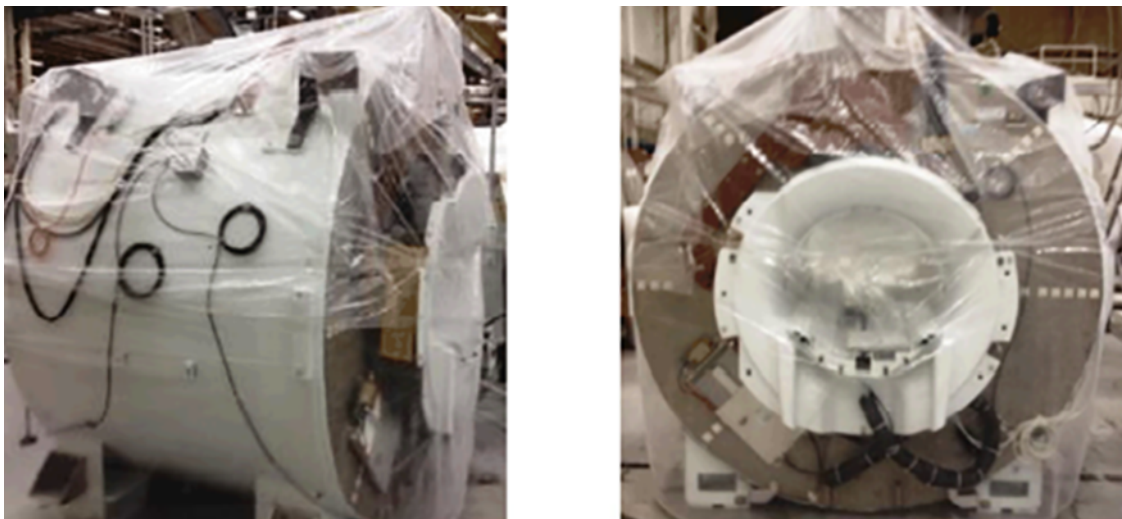
- 4. If applicable, coil cables and hoses and put them inside the bore.

Figure 6-20 Typical cable wrapping and orientation at the magnet rear



- 5. If present, protect cables and hoses with bubble wrap and put them in the bore.
- 6. Bubble wrap any parts that protrude from the side of the magnet. Use stretch wrap to wrap the magnet from the bottom to middle section.

Figure 6-21 Magnet with stretch wrap



6.5 Installing the lifting rails

Important

This procedure must be done by trained GE HealthCare personnel, a qualified rigger, a third-party vendor, and/or under the supervision of the GE HealthCare asset management/PMI team.

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
Any market available torque wrench with valid calibration	1	-	-

Tools and test equipment			
Item	Quantity	Part number	Manufacturer
3/4 inch Combination Wrench	2	-	-
1-1/8 inch Combination Wrench	2	-	-
19 mm Combination Wrench	2	-	-
Portable Drill Motor (as needed, for use with the drill bits listed in the Consumables table)	1	-	-

Consumables			
Item	Quantity	Part number	Manufacturer
Never-Seez Regular Grade Lubricating and Anti-Seize Compound (or equivalent anti-seize compound)	1	46-294151P8 (or buy from local market)	Bostik
Black Sharpie Pen	1	-	-
14.5 mm (9/16 inch) Diameter Drill Bit (high-speed steel or equivalent)	As needed	-	-
20.5 mm (13/16 inch) Diameter Drill Bit (high-speed steel or equivalent)	As needed	-	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
(For HM, R/CR/RA/YR, and RD series magnets) Lifting Rail FRU Kit: LCC/450/450w magnet (R, RD, CR, HM)	1 of these	5926419	-
(For UA and W/WB series magnets) Lifting Rail FRU Kit: G3/750w magnet (W, WB, UA)		5926420	-
(For AR and PM/PC series magnets) Lifting Rail FRU Kit: IPM/Starter magnet (PM, AR)		5926421	-
(For HM, R/CR/RA/YR, and RD series magnets) Hardware Collector Kit, LCC, LCCW, and DVW	1 of these (Lifting Rail Mounting Bolt Kits)	5926709	-
(For UA and W/WB series magnets) Hardware Collector Kit, 3TLC and Corona		5926710	-

Replacement parts			
Item	Quantity	Part number	Manufacturer
(For AR and PM/PC series magnets) Hardware Collector Kit, PM and AR		5926711	-

Hardware Collector Kit, LCC, LCCW, and DVW (5926709)			
Item	Quantity	Part number	Manufacturer
0.50-13UNC x 2.00 Long, Grade 8, Yellow Zinc, Fully Threaded, Hex Head Screw, ASTM F1941	16	5924341	-
0.50-13UNC, Yellow Zinc, Lock Washer, ASTM F1941	30	5924341-80	-
0.50-13, Yellow Zinc, Flat Washer, ASTM F1941	30	5924341-100	-
0.50-13UNC, Grade 8, Yellow Zinc, Hex Head Nut, ASTM F1941	16	5924341-40	-
Bolt Torque Laminated Placard	1	5926638	-

Hardware Collector Kit, 3TLC and Corona (5926710)			
Item	Quantity	Part number	Manufacturer
0.75-16UNF x 2.50 Long, Grade 8, Yellow Zinc, Fully Threaded, Hex Head Screw, ASTM F1941	20	5924341-10	-
0.75-16UNF, Yellow Zinc, Lock Washer, ASTM F1941	40	5924341-90	-
0.75-16, Yellow Zinc, Flat Washer, ASTM F1941	40	5924341-110	-
0.75-16UNF, Grade 8, Yellow Zinc, Hex Head Nut, ASTM F1941	20	5924341-50	-
Bolt Torque Laminated Placard	1	5926638	-

Hardware Collector Kit, PM and AR (5926711)			
Item	Quantity	Part number	Manufacturer
M12-1.75 x 65 mm Long, Class 10.9, Yellow Zinc, Fully Threaded, Hex Head Screw, ISO 4042	20	5924341-30	-
0.50-13UNC, Yellow Zinc, Lock Washer, ASTM F1941	40	5924341-80	-
M12, Yellow Zinc, Flat Washer, ASTM F1941	40	5924341-120	-

Hardware Collector Kit, PM and AR (5926711)			
Item	Quantity	Part number	Manufacturer
M12-1.75, Class 10, Yellow Zinc, Hex Head Nut, ISO 4042	20	5924341-70	-
Bolt Torque Laminated Placard	1	5926638	-

Safety	
 CAUTION	
 HEAVY EQUIPMENT Two people are required to lift the magnet lifting rails.	
 WARNING	RISK OF DEATH, PERSONAL INJURY, OR EQUIPMENT DAMAGE The use of used and/or damaged bolts, lock washers, flat washers, and nuts could result in equipment and/or component damage, death, or serious physical injury! Use new, undamaged mounting hardware.

In this procedure you will install and torque the lifting rail bolts.

- Use new bolts, lock washers, flat washers, and nuts from the following FRUs, ordered from GE HealthCare.
 - (For HM, R/CR/RA/YR, and RD series magnets)** Hardware Collector Kit, LCC, LCCW, and DVW (FRU 5926709)
 - (For UA and W/WB series magnets)** Hardware Collector Kit, 3TLC and Corona (FRU 5926710)
 - (For AR and PM/PC series magnets)** Hardware Collector Kit, PM and AR (FRU 5926711)
- Inspect the new bolts, lock washers, flat washers, and nuts. Make sure that the threads of the bolts are not stripped or damaged. Use each bolt and insert by hand to make sure there is no binding (bolts must be inserted by hand or with a hand tool only, no power tools). Do not use any damaged hardware.

NOTE

Visually examine each tapped hole in the magnet feet. If any threaded bolt hole is damaged, you must drill out the current threaded hole for clearance of the bolt as shown in the following table. The bolts supplied in the FRU kits are longer than the original and will allow for a nut to be used (drill bits are procured locally).

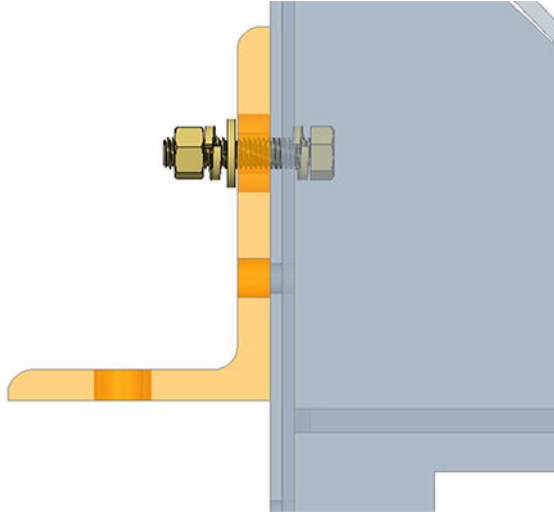
Bolt size	Magnet type (serial number)	Drill size
1/2 inch	LCC/450/450W magnet (R, RD, CR, HM, RA, YR)	14.5 mm (9/16 inch)
M12	IPM/Starter magnet (PM, PC, AR)	14.5 mm (9/16 inch)
3/4 inch	3TLC/G3 magnet (UA, W, WB)	20.5 mm (13/16 inch)

NOTE

If any parts are damaged, order new parts from the tables shown above.

3. Use the bolts, lock washers, flat washers, and nuts to secure each lifting rail. Refer to the figure below for the correct washer sequence.

Figure 6-22 Hardware sequence



4. Before installation, apply Never-Seez (or equivalent anti-seize compound) to the external threads of each bolt, starting at the bottom of the bolt (from thread end toward the bolt head).

5.

WARNING



RISK OF DEATH, PERSONAL INJURY, OR EQUIPMENT DAMAGE

Overtorquing the nuts or bolts could result in equipment and/or component damage, death, or serious physical injury!

Do not overtighten the nuts or bolts.

Install the bolts and lock washers from the inside of the magnet foot as shown in the figure below. Using a hand tool, tighten the bolt until the lock washer is fully compressed, but do not exceed 90 ft-lbs (122 N m). Install and tighten all bolts before installing the lifting rail.

Figure 6-23 Installed bolts

**NOTE**

Do not use powered torque drivers. Powered drivers may supply higher torque that can damage the bolts.

6.

WARNING**RISK OF DEATH, PERSONAL INJURY, OR EQUIPMENT DAMAGE**

Overtorquing the nuts or bolts could result in equipment and/or component damage, death, or serious physical injury!

Do not overtighten the nuts or bolts.

Put the orange lifting rail over the bolts and as far as possible to the rear of the magnet. Then install the flat washers, lock washers, and nuts. Torque the nuts to 90 ft-lbs (122 N m). If any bolt holes on a magnet foot were drilled out then the head of the bolt will need to be held while torquing the nut.

NOTE

Do not overtighten the nuts.

Figure 6-24 Orange lifting rail in position (left) and with hardware installed (right)

**NOTE**

Digitized Per Pin (DPP) cables are located near the bottom of the magnet. Use caution when mounting the rails so that cables are not pinched or damaged.

Figure 6-25 Cables hanging on lifting rail



7. Visually do a check of each bolt to make sure both the inner and outer lock washers are compressed and not loose and the rails have no movement.
 - If any lock washer is not compressed, repeat the previous installation steps.
 - If any lock washer will not compress but the bolt has been tightened to the correct torque value, stop and do not continue. Contact GE HealthCare for assistance.
8. Make sure the rails are installed against the magnet feet at all four locations before moving.

9. Use a permanent marker to add a witness mark on each bolt and on the rail.

Figure 6-26 Example of a witness mark



6.6 Removing life support

In this procedure you will disconnect the compressor and Magnet Monitor.

1. If the adsorber, helium compressor lines, or compressor cryocooler components are being sent to salvage, do the following substeps:
 - a. Refer to the appropriate manual below for your magnet series, and depressurize the components for disposal.

Magnet series	Manual title	Part number	Procedure
AR	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
HM	Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
PM or PC	Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
R, CR, RA, or YR	Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
RD	Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
UA	Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal
W or WB	Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN	Replacement and Maintenance > Depressurizing compressor components for disposal

- b. Remove the Magnet Monitor from the wall and package it with all cables in a box. Make sure to ship this box with the magnet.
 - c. Package all compressor hoses and cables with the compressor. Make sure to ship with the magnet.

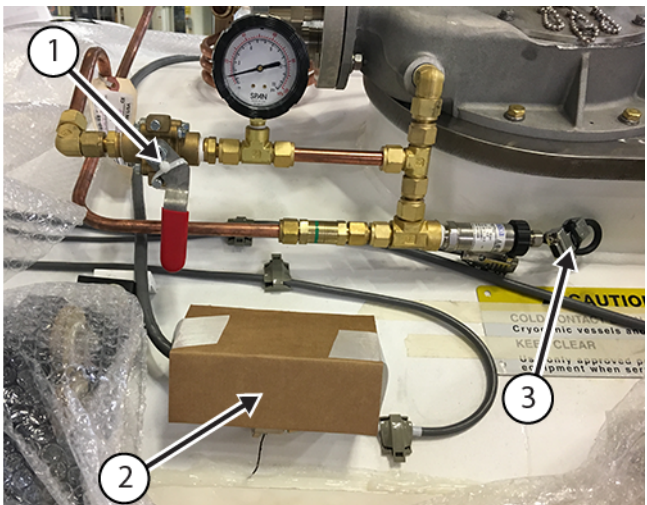
2. Make sure both V2 and V3 are closed for domestic shipping and secure with cable ties.

Figure 6-27 V3 valve in the closed position



3. Make sure fill ports and ramp ports are tightened. If not, use a strap wrench (or similar type of wrench) to tighten the fill and ramp ports.
4. Put a cardboard box over the instrumentation connector assembly.

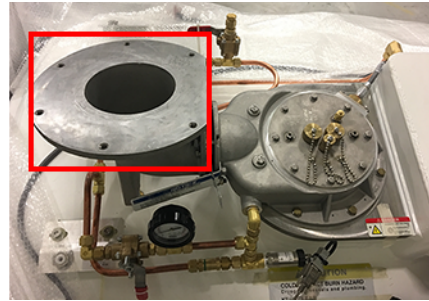
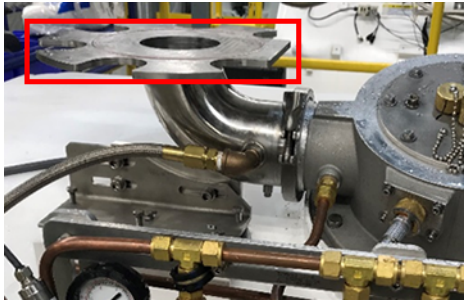
Figure 6-28 Cardboard box over the instrumentation connector assembly



1	V2 valve tied
2	Cardboard box over instrumentation connector assembly
3	Pigtail secure to transducer

5. **(For domestic shipments)** Put cardboard over the open top of the 90-degree elbow and secure the cardboard with tape.

Figure 6-29 Open top of the 90-degree elbow - AR/PM/PC magnet (left) and legacy magnet (right)



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Chapter 7 Moving and transporting the magnet

Important

The procedures in this chapter must be done by a third-party vendor, GE HealthCare asset management team, and/or GE HealthCare PMI.

7.1 Handling the magnet

1. Record all items which will be moved or stored. Send the list of items with the magnet and keep a copy with the FE, PMI, or asset team.
2. Order any missing or damaged items to make sure the magnet can be connected to the compressor and monitored for life support.

For example, you may need to order Core Kit 2350809, which contains parts that can be used to replace missing or damaged Magnet Monitor hardware.

3. Move the magnet in conformance with the appropriate *Magnet Handling Manual* for your magnet series.

Magnet series	Manual title	Part number
AR	Magnet Handling 3.0T AR Series Magnets	5869669-1EN
HM	Magnet Handling 1.5T HM Series Magnets	5836617-1EN
PM or PC	Magnet Handling 1.5T PM Series Magnets	5807458-1EN
R, CR, RA, or YR	Magnet Handling 1.5T R Series Magnets	5836628-1EN
RD	Magnet Handling 1.5T RD Series Magnets	5836619-1EN
UA	Magnet Handling 3.0T UA Series Magnets	5836648-1EN
W or WB	Magnet Handling 3.0T W-WB Series Magnets	5836649-1EN

7.2 Storing the magnet at the warehouse

The steps below are for monitoring MR magnets that are stored at a warehouse while waiting to be shipped to their final destination. The Magnet Monitor, compressor, and cables necessary for the procedure should come with the magnet.

For more detailed information about the steps below, refer to the *Magmon3 Installation and Service Manual* (5124576) or *Magnet Monitor 4 Operating and Service Manual* (5804162) and the appropriate *Magnet Handling Manual* for your magnet series.

1. Make sure there is an internet connection, which is necessary to install and monitor the Magnet Monitor. With an internet connection and correct configuration, the Magnet Monitor will send data to the back office and the magnet can be monitored remotely.

The Magnet Monitor is connected to the magnet, and will show an alarm when any parameter is out of specification at the time an event occurs on the magnet. It monitors the cryogen level inside the magnet and controls the internal helium vessel pressure. The Magnet Monitor connection also measures many physical parameters of the magnet. After the compressor is attached and working, the magnet does not lose helium.

NOTE

If there is no internet, the helium level and pressure of the magnet must be checked two or three times daily and the values recorded in the form below.

If monitoring is done by a third party, the third party must email the completed form to GE HealthCare (the FE, PMI, or asset team) or the Licensed Channel Partner.

Figure 7-1 Sample form

MR Helium check list								
#	Date	Time	Serial number of the MR	Helium level in %	Pressure	Comments	Check by	Signature
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								

2. The GE HealthCare team must make sure that the list of items with the magnet (see [7.1 Handling the magnet on page 59](#)) are still with the magnet. The team must order any missing or damaged items to make sure the magnet can be connected to the compressor and monitored for life support.

Make sure the following items are on the list:

- Magnet Monitor Unit (MM3 or MM4)
 - Compressor (hoses and cables)
 - Any other cables necessary to connect the Magnet Monitor and compressor to the magnet
3. GE HealthCare or a Licensed Channel Partner must connect the Magnet Monitor and compressor per the standard installation process.
 - a. Get power sources for both the Magnet Monitor and the compressor.
 - b. Make sure there is correct cooling per compressor type (for example, water versus air cooled).
 - c. Connect the Magnet Monitor to the back office (internet) if possible. If there is no connection, see [Step 1](#) for more information about the manual form and checks that must be completed daily.
 4. After installation in the warehouse is completed, set the reference pressure level for magnet pressure to 1.0 psi (0.9 to 1.1 psi operating pressure limits).
 5. After the compressor is connected, the pressure should decrease. When the pressure drops below 4 psi, it will stop venting unless the 15.7 psia valve is installed.
 6. If the magnet fails any of the above criteria, contact the local GE HealthCare FE, Licensed Channel Partner, or GE HealthCare Online Center.

Appendix A Reference manuals

Manual title	Part number
Magnet and Cryogen Manual for 3.0T AR Series Magnets	5871810-8EN
Magnet and Cryogen Manual for 1.5T HM Series Magnets	5927844-8EN
Magnet and Cryogen Manual for Platform Magnets	5796143-4EN
Magnet and Cryogen Manual for 1.5T R Series Magnets	5928384-8EN
Magnet and Cryogen Manual for 1.5T RD Series Magnets	5928432-8EN
Magnet and Cryogen Manual for 3.0T UA Series Magnets	5928440-8EN
Magnet and Cryogen Manual for 3.0T W or WB Series Magnets	5928479-8EN
Magnet Handling 3.0T AR Series Magnets	5869669-1EN
Magnet Handling 1.5T HM Series Magnets	5836617-1EN
Magnet Handling 1.5T PM Series Magnets	5807458-1EN
Magnet Handling 1.5T R Series Magnets	5836628-1EN
Magnet Handling 1.5T RD Series Magnets	5836619-1EN
Magnet Handling 3.0T UA Series Magnets	5836648-1EN
Magnet Handling 3.0T W-WB Series Magnets	5836649-1EN
Magmon3 Installation and Service Manual	5124576
Magnet Monitor 4 Operating and Service Manual	5804162
MR Service Safety Manual	5452735

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Appendix B Magnet pressures – P_{MM4} versus psig

Introduction

The AR and PM/PC series magnet design involves the use of new absolute pressure components (pressure transducer, relief valves). These components and equipment help make magnet performance more consistent across install sites of various atmospheric pressures and elevations.

Due to this change in how AR and PM/PC series magnet pressure is regulated and reported to the Magnet Monitor (MM4), some sites may require the application of different MM4 and magnet pressure settings during service procedures that require opening the magnet helium space to the atmosphere. With the same absolute pressure in the magnet, sites at higher elevations will experience larger pressure differences between the magnet and the atmosphere, which will result in more rapid helium gas flow out of the magnet wherever it is opened.

To make sure of the safety of anyone working on AR and PM/PC series magnets, the following information must be considered before doing any service procedures.

Local pressure

Each site must determine their local atmospheric pressure based on their elevation. The [Conversion chart for elevation above sea level to local atmospheric pressure on page 64](#) that follows can be used to help convert from the local elevation above sea level (feet or meters) to local atmospheric pressure (psia).

Pressure conversion

For all service procedures referencing processes in magnet psig, the FE must be aware of the equivalent MM4 pressure (P_{MM4}) before opening the magnet.

The analog gauge on the magnet reads pressure directly in psig relative to the local atmospheric pressure. Any references to psig pressure requirements in a process should match the analog gauge reading at all sites.

MM4 will only provide magnet pressure readings in psig relative to sea level (P_{MM4}). Any pressure control setpoints entered into MM4 must also be in units of psig relative to sea level (P_{MM4}). To convert from psig pressure requirements in the manual to psig readings or entries in MM4 relative to sea level, use the following formula:

$$P_{MM4} = P_{GAUGE} - (14.7 \text{ psia} - P_{AIR})$$

- **P_{MM4}** : Helium Vessel Pressure, read by Magnet Monitor, in psig (relative to atmospheric sea level pressure)
- **P_{GAUGE}** : Process Pressure Requirement, from manual, in psig (relative to local atmospheric pressure)
- **P_{AIR}** : Local Atmospheric Air Pressure, from the [Conversion chart for elevation above sea level to local atmospheric pressure on page 64](#), in psia

Use the formula above to compare the readings between the analog gauge and MM4. If the pressures between the analog gauge and MM4 do not match using this formula, this may indicate the need to replace a gauge or pressure transducer, troubleshoot the connections to MM4, or double-check the local atmospheric pressure assumption. Resolve any pressure reading discrepancies to make sure of correct pressure readings during service procedures.

Negative pressure

Be aware of the local atmospheric pressure while opening the magnet. Keeping the magnet pressure above the local atmospheric pressure will help to avoid cryopumping moisture or creating ice inside of the magnet.

V2 valve pressure check

In order to safely assess the magnet pressure, the V2 valve on the plumbing assembly should be opened before any other locations of the magnet are opened to the atmosphere. Loud helium flow after opening the V2 valve may indicate high magnet pressures, depending on the requirements of the service process being done.

Pressure questions

If there are any questions about magnet pressure or converting pressures between P_{MM4} and P_{GAUGE} , contact your Regional Magnet and Cryogen (MAC) Team Leader or the Magnet Support Team at the Online Center before continuing.

Conversion chart for elevation above sea level to local atmospheric pressure

Elevation above sea level		Local atmospheric pressure, psia
Feet	Meters	
-2,000	-610	15.79
-1,000	-305	15.24
-500	-152	14.96
0	0	14.7
500	152	14.43
1,000	305	14.17
1,500	457	13.92
2,000	610	13.66
2,500	762	13.42
3,000	914	13.17
3,500	1,067	12.93
4,000	1,219	12.69
4,500	1,372	12.46
5,000	1,524	12.23

P_{MM4} versus psig

For the pressure requirements stated in magnet and cryogen procedures, there are two different values that may be referenced, depending on how the number is intended to be used:

- **P_{MM4} :** Pressures related to MM4 values appear in the procedures like the following example: $P_{MM4} = 1.0 \text{ psi}$. These are values that will appear directly as MM4 readings. The pressure requirement within the procedures may result in negative P_{MM4} values needing to be entered into MM4 due to local elevation. Some early MM4 software versions may not accept negative setpoints, in which case it is okay to enter a set of low-positive values (such as 0.1 psi to 0.2 psi) in place of the negative values.

- **P_{GAUGE}**: Pressures related to local gauge pressure appear in the procedures like the following example: 2.5 *psig*. These are values that will appear directly on the analog gauge, and typically apply to processes where the magnet must be opened to atmosphere.

If the safety value after calculations is negative, set MM4 to the lowest value and continue to vent down with the V2 valve until you reach the safety level to open the magnet.

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Revision history

Revision	Date	Description
		Controlled document for English is posted as DOC2698307.

Revision	Date	Description
4	May 2023	Updated the following sections: • Updated references to <i>GE HealthCare</i>. • Changed <i>non-Platform</i> references to <i>legacy</i>. Changed <i>Platform</i> references to <i>AR and PM/PC</i>. • Replaced all instances of 5924907-XXX¹ with 5924907-1EN, and 5918186-XXX¹ with 5918186-1EN. Removed the corresponding footnote statement ¹<i>XXX is the language indicator of the manual</i>. • Changed references to <i>Magnet and Cryogen Manual for Passively Shimmed Magnets (5495018)</i> and <i>Magnet and Cryogen Manual for Actively Shimmed Magnets (5495019)</i> to the appropriate new series-based <i>Magnet and Cryogen Manuals</i>. • Anywhere the PM series magnet is mentioned, added the PC series. • Added a statement to the following sections about shutting down and removing 7T magnets: ◦ 1.1 Purpose ◦ 1.3 Scope • 1.1 Purpose, updated the <i>Magnet deinstallation workflow</i>. • 1.6 Roles and responsibilities, added Table 1-2 Deinstallation functions and responsibilities. • In 2.2 Safety training, removed the <i>Superconducting Magnets, Introduction</i> section. The temperatures listed were not applicable to deinstallation. • Chapter 3 Preparing the site: ◦ In step 6, added mention of the NPT plug. ◦ Updated the title for Figure 3-1 (previously 3-2) to <i>AR/PM/PC V2 and V3 vent valves</i>, and added an additional callout for V2. ◦ Updated the title for Figure 3-2 (previously 3-1) to <i>Legacy V2 vent valve and pressure gauge</i>. ◦ Added <i>Figure 3-3 Legacy V3 vent valve</i>. ◦ Added <i>Figure 3-4 Legacy NPT plug</i>. • In 3.5.3.2 Potential injury risk, updated the Notice to reference a specific section name. • Added steps 3 through 5 to 4.3.3 Ramping down with the auxiliary cable due to SavCon or shim lead damage (only R, CR, RA, YR, W, and WB series). • 6.1.1.1 Converting an air or ocean shipment (legacy): ◦ Updated the title for Figure 6-1 to <i>Legacy V2 vent valve and pressure gauge</i>. ◦ Added step 6, to turn off the compressor. ◦ Added step 11, to turn on the compressor. • 6.1.2.1 Converting an air or ocean shipment (AR/PM/PC): ◦ In the <i>Replacement parts</i> table and step 6, added the new burst disc clamp, 5885378. ◦ Added step 2, to turn off the compressor. ◦ In step 6, changed 5797712 to FRU 5807233. ◦ In step 7, improved the instruction on how to "fold" the O-ring. ◦ Added step 10, to turn on the compressor.

Revision	Date	Description
		◦ If the metal tab on a burst disc breaks during removal, the burst disc no longer has to be discarded. The Important note in step 11 was updated to reflect this change. • In 6.3 Unfixing the magnet from the floor, added step 4, about removing magnet covers. • 6.4 Packaging and protecting the magnet: ◦ Added a note before step 1. ◦ Added a sentence to step 1 about when to cut the hose. • 6.5 Installing the lifting rails: ◦ Updated the important note about who can do the procedure. ◦ In step 5, added clarification to "not exceed 90 ft-lbs (122 N m)." ◦ Added a warning about overtorquing to step 6. • In 6.6 Removing life support, removed step 2a (If the magnet is being shipped out the same day, make sure there is a cable tie securing the V3 valve in the open position...) and corresponding <i>Figure 6-27 V3 valve in the open position</i>. Changed step 2b to step 3. • Chapter 7 Moving and transporting the magnet, updated the note. Changed "and" to "and/or". • In step 1 of the following sections, changed PM to PMI: ◦ 7.1 Handling the magnet ◦ 7.2 Storing the magnet at the warehouse • In 7.1 Handling the magnet, updated step 2 to include an example of what to order. • Appendix B Magnet pressures – P_{MM4} versus psig: In the P_{MM4} versus psig section, updated how to enter negative setpoints in early MM4 software versions.
3	August 2022	Updated the following sections: • 3.5.3.2 Potential injury risk, added two bullet points to the "NEVER do" list: ◦ Reuse mounting hardware when lifting the magnet ◦ Overtighten nuts or bolts above the recommended torque specification • 4.3 Removing the field from the magnet, corrected a typo of the word "shipment" in: ◦ 4.3.1 Normal ramp down, step 3 ◦ 4.3.2 Ramping down with the auxiliary cable due to significant icing in the vertical penetration (only R, CR, RA, YR, W, and WB series), step 6 ◦ 4.3.3 Ramping down with the auxiliary cable due to SavCon or shim lead damage (only R, CR, RA, YR, W, and WB series), step 4 • 6.5 Installing the lifting rails: ◦ Added a danger about the use of used and/or damaged hardware to the Safety table. ◦ Added a danger about the risk of overtorquing the nuts or bolts before step 5.

Revision	Date	Description
2	July 2022	Added the following sections: 1.2 Returning used systems - All sections in Chapter 3, Preparing the site 4.3 Removing the field from the magnet 6.4 Packaging and protecting the magnet Updated the following sections: - On the cover page, changed the title from <i>SIGNA Hero System Deinstallation Manual</i> to <i>Magnet Deinstallation Manual</i>. - The Language Policy was updated to list the languages into which the manual is translated. - Changed references of "5918186-1EN" to "5918186-XXX¹" with the addition of a footnote, "¹XXX is the language indicator of the manual." - In Chapter 1 Introduction: - Updated 1.1 Purpose: - Changed manual title referenced from <i>SIGNA Hero System Deinstallation Manual</i> to <i>Magnet Deinstallation Manual</i>. - Added a reference to the <i>MR System Deinstallation Manual</i>. - Changed references of "SIGNA Hero system magnet" to "GE Healthcare system magnet". - Added MM3 to example list of harvested parts. - Added a deinstallation workflow. - Updated 1.3 Scope. Removed reference to SIGNA Hero system magnets. Listed all magnets this manual applies to, as well as those it does not. - In 1.4, Definition of terms and acronyms, added PMI. - In 1.5 References, changed manual title referenced from <i>SIGNA Hero System Deinstallation Manual</i> to <i>Magnet Deinstallation Manual</i>. - In 1.6 Roles and responsibilities, for the project manager role, modified the last sentence into two statements. Spelled out the acronym MSL; removed the acronyms HPM, UMI, and ASM; and added the acronym PMI. - In Chapter 2 Safety, updated 2.2 Safety training. - Added a bulleted list of three necessary trainings. - In the "Emergency Rampdown of Magnetic Field" section, replaced the asphyxiation hazard statement with more specific details as stated in the <i>MR Service Safety Manual</i> (5452735). - In Chapter 3 Preparing the site, added steps 7 and 8 to examine the quench pipe. - In Chapter 4 Shutting down the system: - In 4.1 Archiving patient images, replaced existing steps with a single step referring to the <i>MR System Deinstallation Manual</i>. - In 4.2 Ramping down a magnet to zero field, updated the table to include HM, PM, R, CR, RA, YR, RD, W, and WB manual references. Also removed the note about this procedure qualifying the magnet for ground transport only, as the addition of section 4.3, Removing the field from the magnet, allows for all transport types.

Revision	Date	Description
		◦ In 4.4 Disconnecting the helium vent pipe, steps 1 and 5, updated the table to include HM, PM, R, CR, RA, YR, RD, W, and WB manual references. Added a figure to step 6a. In step 6b, changed "90-degree elbow" to "any vent pipe 90-degree elbows" and added a figure for clarification. ◦ In 4.5 Cryogen and magnet life support, added MM3 references. • Former section 6.3 Shipping the magnet to salvage is now Chapter 5 Preparing the magnet for salvage. In the opening paragraph, the phrase "and to keep vacuum integrity" was removed and "cryogen" was changed to be plural. • In Chapter 6 Preparing the magnet for shipment: ◦ In section 6.1 Converting a magnet to shipment configuration: • Added reference to air and ocean shipment. • Added a Safety table. • Replaced 5.1.1 Configuring an AR or UA series magnet for domestic shipment with: • 6.1.1 Configuring a non-Platform magnet for shipment • 6.1.2 Configuring a Platform magnet for shipment • In 6.1.3 Doing a leak check, updated the table to include HM, PM, R, CR, RA, YR, RD, W, and WB manual references. ◦ Updated 6.2: • Title changed from "Removing the table and dock" to "Removing the other magnet components". • "Removing the coolant water" changed to "Removing the coolant from magnet components". • Procedures "Removing the dock" and "Removing the patient table" combined into one list item, and listed first. ◦ Updated 6.5 Installing the lifting rails: • In the Important note, changed "Field Engineer or" to "personnel and/or"; changed "rigger" to "qualified rigger". • Removed the lifting rails from the Tools and test equipment table; they are included in the new rail FRU kits (5926419, 5926420, and 5926421) added to the Replacement parts table. Updated the torque wrench description to no longer include a specific torque value. Added the 3/4 inch Combination Wrench, 1-1/8 inch Combination Wrench, 19 mm Combination Wrench, and Portable Drill Motor. • In the Consumables table, replaced Loctite #242 and #243 with Never-Seez Regular Grade Lubricating and Anti-Seize Compound (46294151P8). Added the 14.5 mm Diameter Drill Bit and 20.5 mm Diameter Drill Bit. • In the Replacement parts table, added 5926709; replaced 5887267 with 5926710; and replaced 5887268 with 5926711. Updated the descriptions for the lifting rail mounting bolt kits to clarify the magnet series. • Added a Hardware table for 5926709. Replaced the Hardware table for 5887267 with a table for 5926710. Replaced the Hardware table for 5887268 with a table for 5926711. • Inserted a new step 1 to clarify "using the new bolts...ordered from GE Healthcare" and listed the FRUs.

Revision	Date	Description
		• In step 2, added nuts to the list of hardware mentioned. Added the text "... (bolts must be inserted by hand or with hand tool only, no power tools). Do not use any damaged hardware." A note was also added about drilling damaged threaded bolt holes in the magnet feet. • In step 3, added nuts to the list of hardware mentioned, and updated the figure. • Removed step about cleaning all hardware surfaces with a cleaning solvent. • Updated step 4 to clarify "Before installation"; to use Never-Seez instead of Loctite; and to clarify the direction of the application. • Previous step 2, about aligning the rails, and step 6, about tightening bolts using hand tools and a torque wrench, were reconstructed into a new step 5, about installing the bolts and lock washers from the inside of the magnet foot and tightening bolts by hand, and step 6, about placing the rails, installing hardware, and torquing nuts. • Added step 7 to visually do a check of the tightness of each bolt. • In step 8, changed "installed on the magnet feet" to "installed against the magnet feet". • In step 9, updated the figure to show witness marks on the rail. ◦ Updated 6.6 Removing life support: • Updated the table in step 1 to include HM, PM, R, CR, RA, YR, RD, W, and WB manual references. • Removed the "V2 vent valve and pressure gauge" and "V3 vent valve" figures from step 2. Added substeps further clarifying the configuration. • Clarified that step 4 applies to domestic shipments, and updated the figure to show an example of a non-Platform magnet. • In Chapter 7 Moving and transporting the magnet: ◦ In 7.1 Handling the magnet, updated the table in step 3 to include HM, PM, R, CR, RA, YR, RD, W, and WB manual references. ◦ In 7.2 Storing the magnet at the warehouse, added MM3 references to the introduction and step 2. • In Appendix A Reference manuals, added additional <i>Magnet and Cryogen Manual</i> and <i>Magnet Handling Manual</i> references for HM, PM, R, CR, RA, YR, RD, W, and WB series. Also added the <i>Magmon3 Installation and Service Manual</i>. Removed <i>Customer Site Storage Requirements</i> (5182674) since its contents is not referenced in this manual. • In Appendix B Magnet pressures – P_{MM4} versus psig, everywhere AR is mentioned, added "and PM".
1	April 2022	Initial release.



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